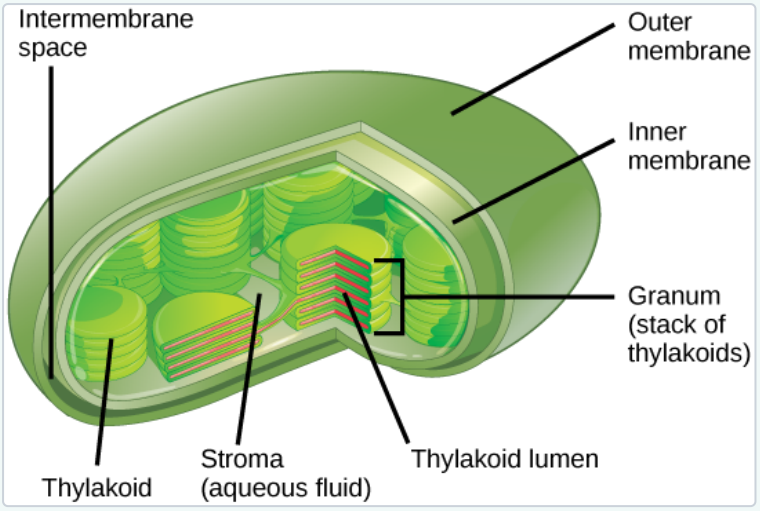
PHOTOSYNTHESIS

WHERE IT HAPPENS · THE CHLOROPLAST



Cut-away view of a chloroplast.
Image: OpenStax, *Biology*, via Wikimedia Commons, CC BY 4.0.

- ✔ Thylakoid membrane: **light reactions** (ATP, NADPH, O_2)
- ✔ Stroma: **Calvin cycle** (CO_2 → sugar)
- ✔ Granum = stack of thylakoids: more membrane, more light capture

OVERALL EQUATION · LIGHT ENERGY

$$6\,CO_2 + 6\,H_2O \longrightarrow C_6H_{12}O_6 + 6\,O_2$$

THE TWO STAGES

CRITERION	LIGHT REACTIONS	CALVIN CYCLE
Where	thylakoid membrane	stroma
Needs	light, H_2O	CO_2, ATP, NADPH
Makes	ATP, NADPH, O_2	G3P (→ glucose)
Key point	O_2 comes from **water**	RuBisCO fixes CO_2

ENERGY FLOW

LIGHT absorbed by chlorophyll → **ATP + NADPH** chemical energy → **SUGAR** CO_2 fixed

! KEY POINT

The light reactions turn light into **chemical energy** (ATP, NADPH); the Calvin cycle **spends** it to build sugar from CO_2.

MUST KNOW

MEMORIZE

- ✔ Chlorophyll absorbs mostly **blue** and **red**; reflects green.
- ✔ Water splitting (photolysis) releases O_2.
- ✔ Limiting factors: light, CO_2, temperature.



✘ COMMON TRAP

× "The Calvin cycle only happens in the dark." → **It does not need light directly, but depends on ATP/NADPH from the light reactions.**

× "Plants photosynthesize instead of respiring." → **Plants do both.**

★ TIP

Follow the atoms: the C and O in glucose come from **CO_2**; the O_2 released comes from **H_2O**.

References: OpenStax Biology 2e, 8.1 Overview of Photosynthesis; 8.2 The Light-Dependent Reaction; 8.3 Using Light Energy to Make Organic Molecules.